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Reliability in intensive longitudinal studies: A state-space vs. linear mixed model approach

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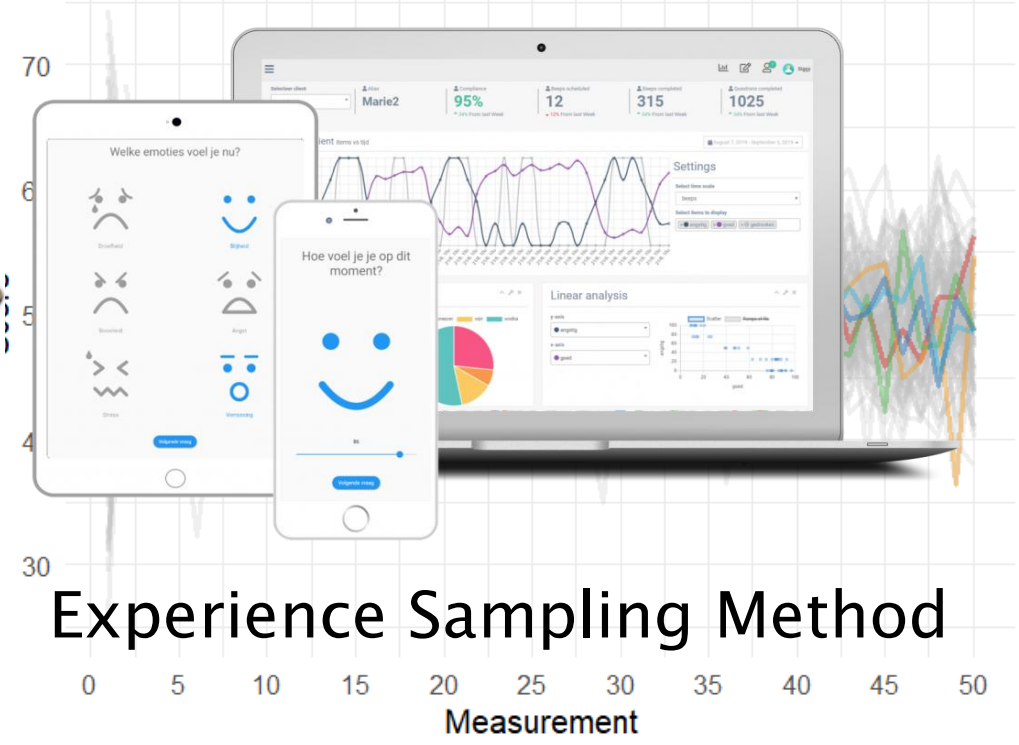
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Intensive Longitudinal Data (ILD)

- ILD is a rapidly growing area in psychological study and social sciences focused on understanding processes and behaviors that unfold over time.
- These studies are characterized by
  - high temporal resolution ⇒ modeling intraindividual dynamics
  - collecting multiple individuals ⇒ modeling interindividual differences



Daily Diary



Experience Sampling Method



Ambulatory Assessment

Reliability & Classical Test Theory (CTT)

- A **reliable** method or measurement instrument is one that gives consistent results when it is repeated.
- An observed-score Y_{ij} for subject i at occasion j can be decomposed into two parts: a true score T_{ij} and a random measurement error ϵ_{ij}

$$Y_{ij} = T_{ij} + \varepsilon_{ij} \text{ and } \sigma_Y^2 = \sigma_T^2 + \sigma_\varepsilon^2$$

  - where $E[Y_{ij}] = T_{ij}, Cor(\varepsilon_{ij}, \varepsilon_{ij'}) = Cor(\varepsilon_{ij}, T_{ij}) = 0$
- There are two approaches to define reliability coefficient (Lord et al., 1968):

Correlation Approach

(with the parallel assumptions)

$$Cor(Y_{ij}, Y_{ij'}) = \frac{\sigma_T^2}{\sigma_T^2 + \sigma_\epsilon^2}$$



Variance Partitioning Approach

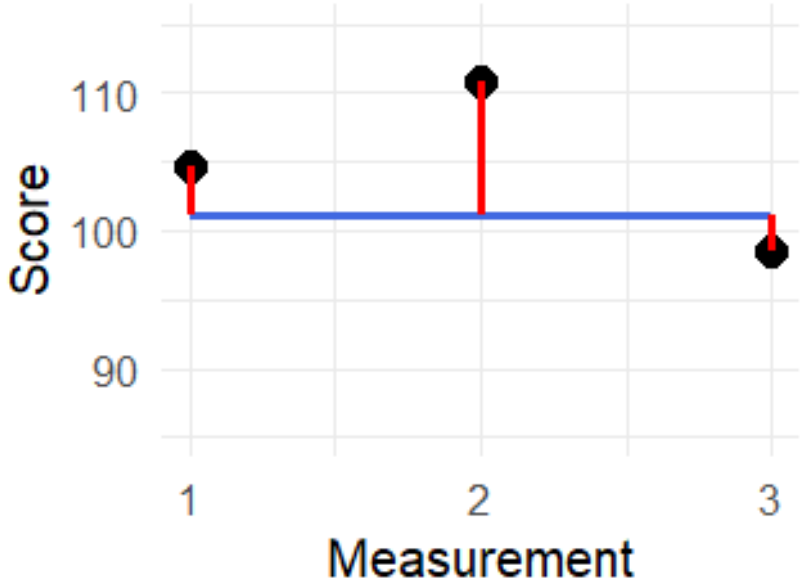
$$\frac{\sigma^2_{\text{Systematic}}}{\sigma^2_{\text{Systematic}} + \sigma^2_{\text{error}}} = \frac{\sigma_T^2}{\sigma_T^2 + \sigma_\epsilon^2}$$

Limitation of CTT

1. Observations in ILD are usually correlated

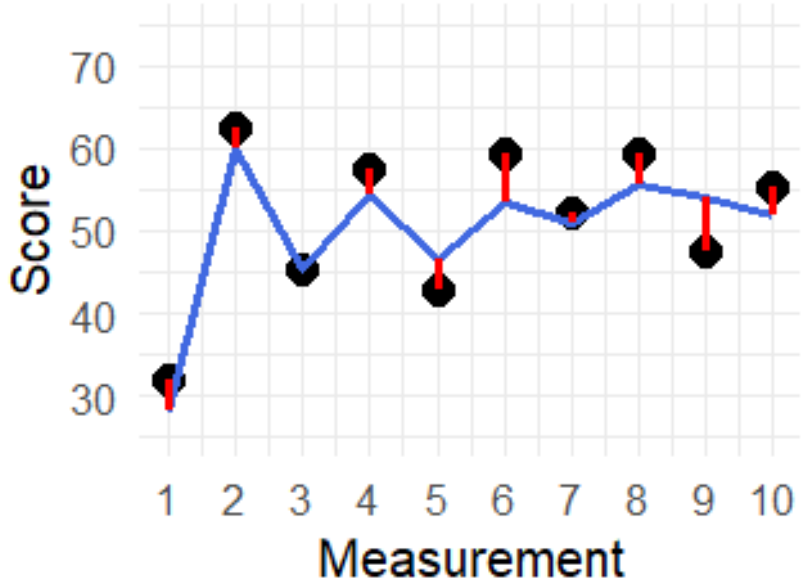
$$Y_t = T + \varepsilon_t$$

Classical Test Theory



$$Y_t = T_t + \varepsilon_t$$

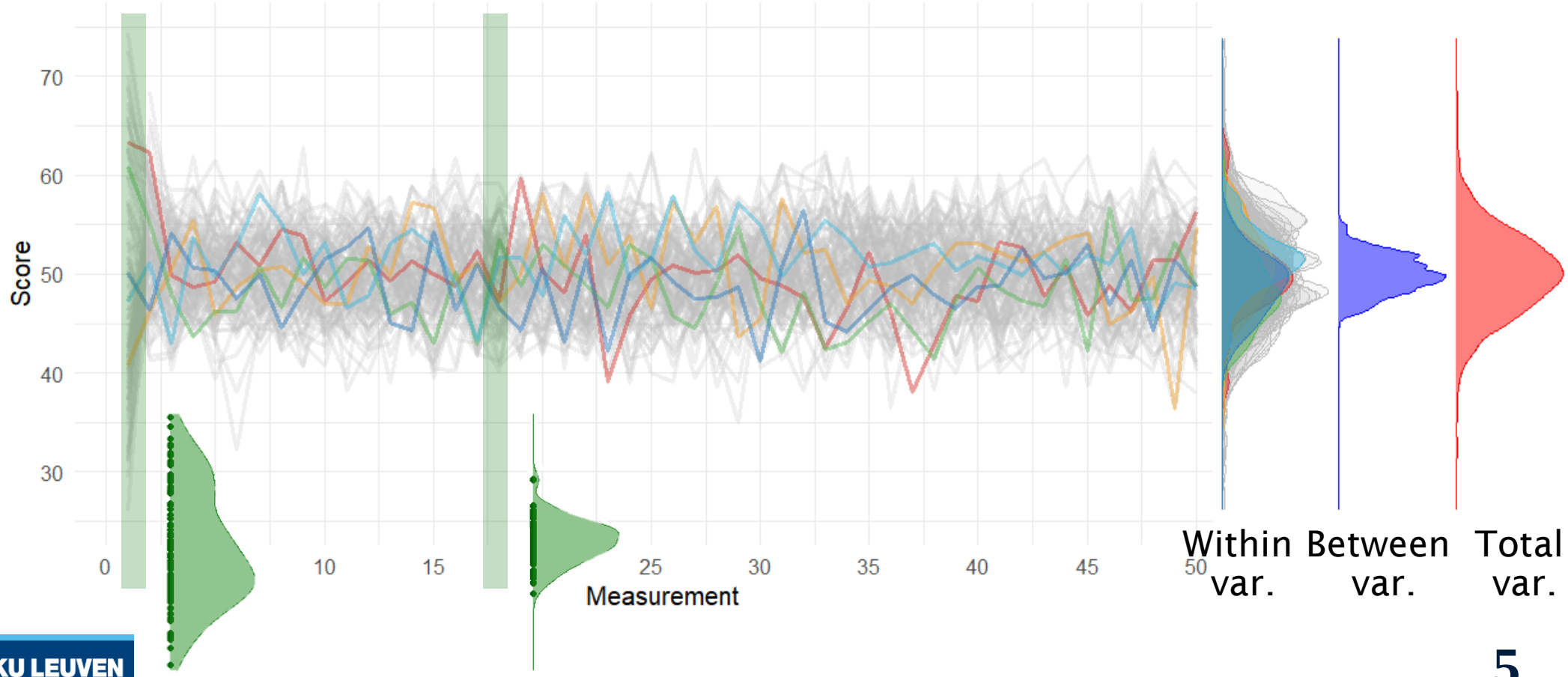
Intensive Longitudinal Data



2. Multiple identifiable sources of factors/errors are not accounted for
   - E.g., gender, item, day effect, etc.

Limitation of CTT

3. ILD are a complicated structure
   - Time-varying variance
   - Nested structure: between-person variance vs. within-person variance



Reliability in Intensive Longitudinal Studies

A special case
- Autoregression process (1)
- Single variable/item

	Linear mixed-effect model with serial correlation	Multilevel state-space model
Correlation approach	Vangeneugden et al. (2004, 2005) LMM-Cor	-
Variance partitioning approach	Laenen et al. (2007, 2009, 2009a) LMM-VP	Schuurman & Hamaker (2019) SSM-VP

Linear Mixed-Effects Models with Serial Correlation

- Let y_{it} be an observed value for person $i = 1, \ldots, N$ at time point $t = 1, \ldots, T$

Fixed effects

Random effects

$b_i \sim N_q(0, \mathbf{D})$

Between person variance

$$\boldsymbol{y}_i = \begin{pmatrix} y_{i1} \\ \vdots \\ y_{iT} \end{pmatrix} = \mathbf{X}_i\boldsymbol{\beta} + \boldsymbol{Z}_i\boldsymbol{b}_i + \boldsymbol{\omega}_i + \boldsymbol{\varepsilon}_i$$

$$\mathbf{V} = Var(\boldsymbol{y}_i) = \mathbf{ZDZ}^\top + \tau^2\mathbf{H} + \boldsymbol{\Sigma}$$

Gaussian stochastic process

$\omega_i \sim N_q(0, \tau^2\mathbf{H})$

E.g., AR(1) serial correlation

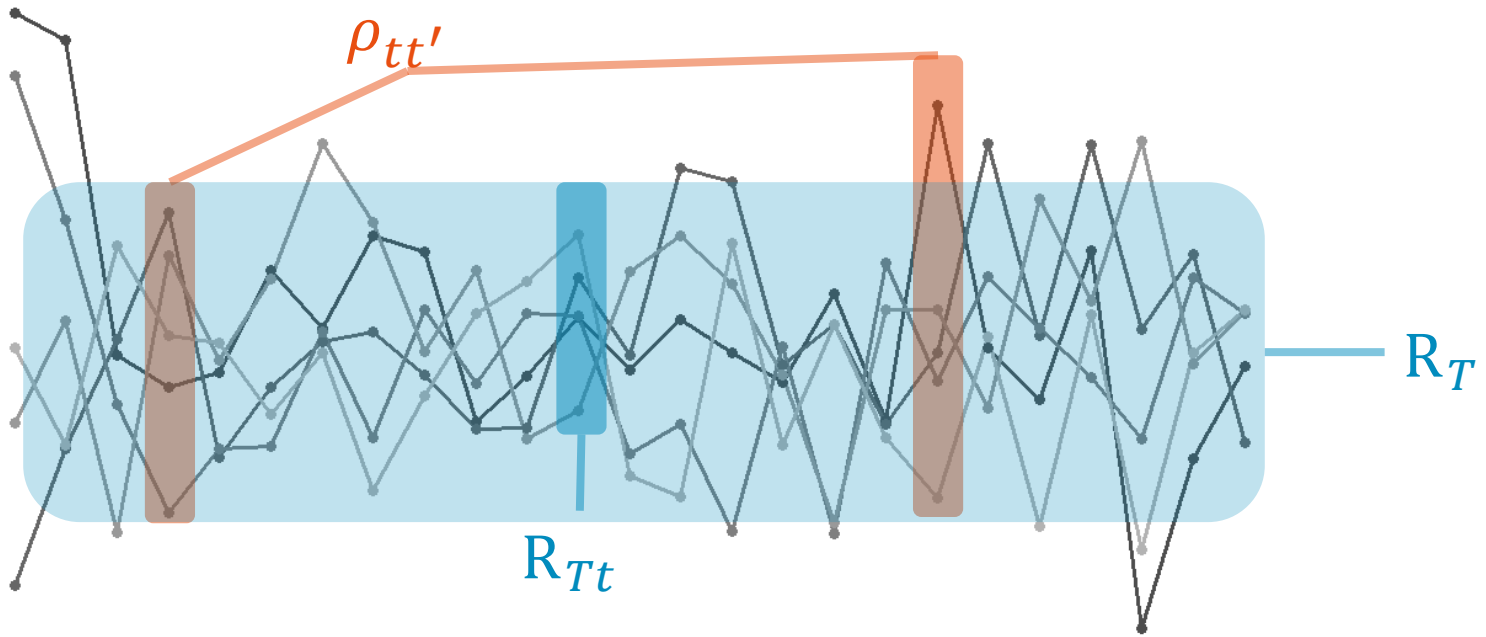
$$\tau^2 = \frac{\sigma_\omega^2}{1-\phi^2},\ \mathbf{H} = \begin{pmatrix} 1 & \rho & \ldots & \rho^{T-1} \\ \rho & 1 & & \\ \vdots & & \ddots & \\ \rho^{T-1} & & & 1 \end{pmatrix}$$

Measurement error

$$\boldsymbol{\epsilon}_i \sim N_T(0, \boldsymbol{\Sigma}),\ \boldsymbol{\Sigma} = \begin{pmatrix} \sigma_1^2 & 0 & \ldots & 0 \\ 0 & \sigma_2^2 & & \\ \vdots & & \ddots & \\ 0 & & & \sigma_T^2 \end{pmatrix}$$

Heterogeneous error variance between time points

Two approaches to Define Reliability



- Correlation approach

$$\rho_{tt'} = Cor(y_{it}, y_{it'})$$

$$= \frac{z_t \mathbf{D} z_{t'}^\top + \tau^2 (\mathbf{H})_{tt'}}{\sqrt{z_t \mathbf{D} z_t^\top + \tau^2 + \sigma_{\epsilon t}^2}\sqrt{z_{t'} \mathbf{D} z_{t'}^\top + \tau^2 + \sigma_{\epsilon t'}^2}}$$

- Variance partitioning approach

Time-varying $$R_{Tt} = \frac{z_t \mathbf{D} z_t^\top}{z_t \mathbf{D} z_t^\top + \tau^2 + \sigma_{\epsilon t}^2}$$

Overall $$R_T = \frac{\mathrm{tr}(\mathbf{Z}\mathbf{D}\mathbf{Z}^\top)}{\mathrm{tr}(\mathbf{Z}\mathbf{D}\mathbf{Z}^\top + \tau^2 \mathbf{H} + \mathbf{\Sigma})}$$

Multilevel State-Space Model

- Multilevel measurement error AR(1) model (Schuurman and Hamaker, 2019)
- Level 1 (within-person)
  - Observation: $y_{it} = \mu_i + \theta_{it} + \varepsilon_{it}, \quad \varepsilon_{it} \sim N(0, \sigma^2_{\varepsilon i})$
  - State: $\theta_{it} = \phi_i \theta_{it-1} + \omega_{it}, \quad \omega_{it} \sim N(0, \sigma^2_{\omega i})$
- Level 2 (between-person)
  - $\begin{pmatrix} \mu_i \\ \phi_i \end{pmatrix} \sim N_2\left(\begin{pmatrix} \gamma_\mu \\ \gamma_\phi \end{pmatrix}, \Psi = \begin{pmatrix} \psi^2_\mu & \psi_{\mu\phi} \\ \psi_{\mu\phi} & \psi^2_\phi \end{pmatrix}\right)$ $\quad \sigma^2_{\varepsilon i} \sim LogN\left(\gamma_{\sigma^2_\varepsilon}, \psi^2_{\sigma^2_\varepsilon}\right)$ $\quad \sigma^2_{\omega i} \sim LogN\left(\gamma_{\sigma^2_\omega}, \psi^2_{\sigma^2_\omega}\right)$

trait | state | measurement error

Auto. coef. | innovation

$$\mathbf{V} = Var(y_{it}) = \psi^2_u + E[\tau^2_i] + \gamma_{\sigma^2_\varepsilon}$$

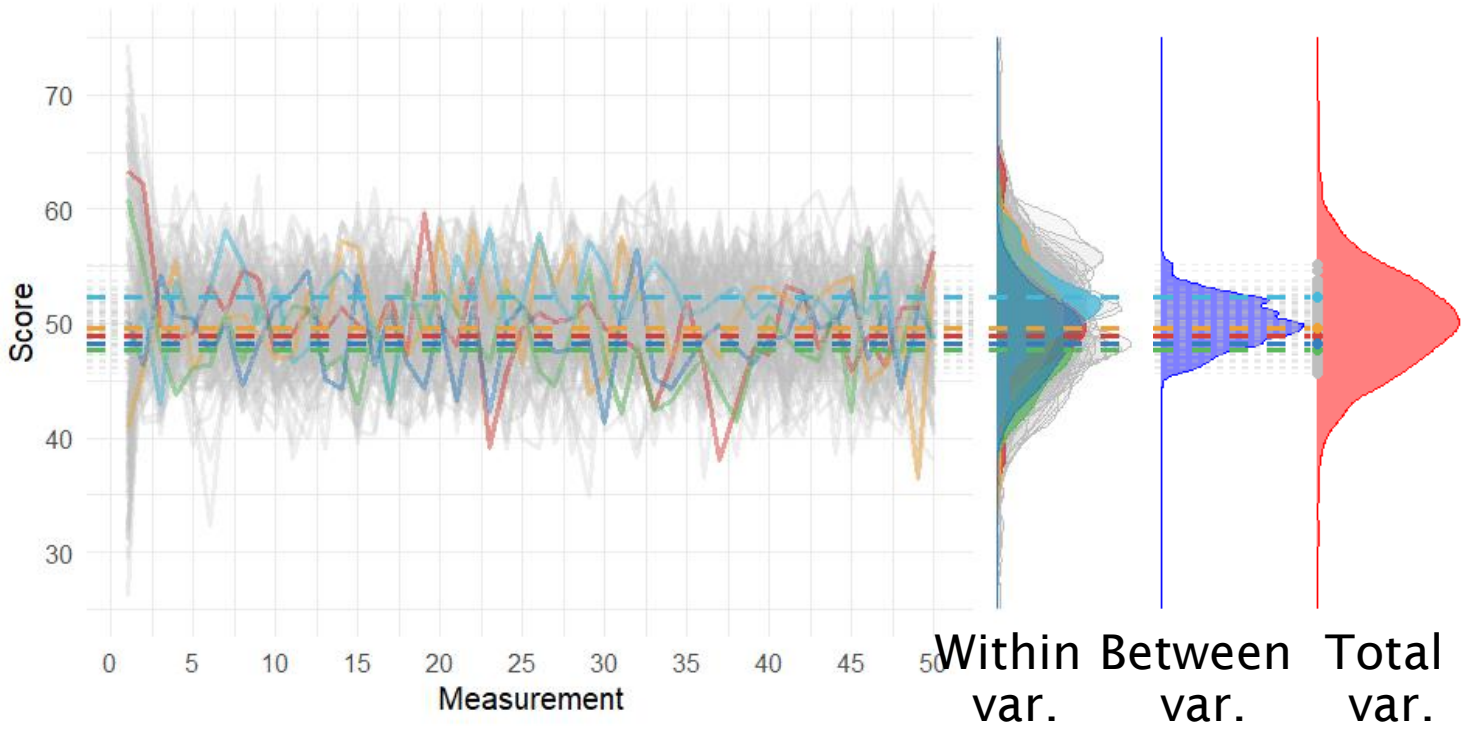
between-person variance (trait score variance)

expected value of within-person variances $\tau^2_i = Var(\theta_{it}) = \sigma^2_{\omega i}/(1-\phi^2_i)$

expected value of measurement error var. $\gamma_{\sigma^2_\varepsilon} = E[\sigma^2_{\varepsilon i}]$

Deriving Reliabilities From the Multilevel SSM

$$y_{it} = \mu_i + \theta_{it} + \varepsilon_{it}$$



Within-person reliability
(for each person)

$$R_i^w = \frac{\tau_i^2}{\tau_i^2 + \sigma_{\varepsilon i}^2}$$

Between-person reliability

$$R^b = \frac{\psi_u^2}{\psi_u^2 + E[\tau_i^2] + \gamma_{\sigma_\varepsilon^2}}$$

Overall Comparison

LMM-Cor	LMM-VP	SSM-Cor	SSM-VP
Heterogeneity between time points		Heterogeneity between people	
$Var(\boldsymbol{y}_i) = \mathbf{ZDZ}^\top + \tau^2\mathbf{H} + \boldsymbol{\Sigma}$		$Var(y_{it}) = \psi_u^2 + E[\tau_i^2] + \gamma_{\sigma_\varepsilon^2}$	
$\rho_{t,t'} = Cor(y_{it}, y_{it'}) = \frac{\psi_\mu^2 + \tau^2\phi^{\lvert t-t'\rvert}}{\sqrt{\psi_\mu^2 + \tau^2 + \sigma_{\varepsilon t}^2}\sqrt{\psi_\mu^2 + \tau^2 + \sigma_{\varepsilon t'}^2}}$	$R_T = \frac{\psi_\mu^2}{\psi_\mu^2 + \tau^2 + \frac{1}{T}\sum_{t=1}^{T}\sigma_{\varepsilon t}^2}$ $R_{Tt} = \frac{\psi_\mu^2}{\psi_\mu^2 + \tau^2 + \sigma_{\varepsilon t}^2}$	$\rho_{t,t'} = Cor(y_{it}, y_{it'}) = \frac{\psi_\mu^2 + E\left[\tau_i^2\phi_i^{\lvert t-t'\rvert}\right]}{\psi_\mu^2 + E[\tau_i^2] + \gamma_{\sigma_\varepsilon^2}}$	$R^b = \frac{\psi_\mu^2}{\psi_\mu^2 + E[\tau_i^2] + \gamma_{\sigma_\varepsilon^2}}$ $R_i^w = \frac{\tau_i^2}{\tau_i^2 + \sigma_{\varepsilon i}^2}$
a decreasing function of the time lag / a full correlation matrix	an overall reliability / time-vary reliabilities	a decreasing function of the time lag	a between-person reliability / within-person reliabilities

Overall Comparison

LMM-Cor	LMM -VP	SSM-Cor	SSM-VP
$\rho_{t,t'}$	R_T R_{Tt}	$\rho_{t,t'}$	R^b R_i^w

Homogeneity between time and people

$\sigma_{\epsilon it}^2 = \sigma_\epsilon^2$, $\sigma_{\epsilon it}^2 = \sigma_\epsilon^2$, $\phi_i = \phi$

Without serial correlation

$\phi = 0$

$$\rho_{t,t'} = \frac{\psi_\mu^2 + \tau^2 \phi^{|t-t'|}}{\psi_\mu^2 + \tau^2 + \sigma_\varepsilon^2} \qquad R_T^b = \frac{\psi_\mu^2}{\psi_\mu^2 + \tau^2 + \sigma_\varepsilon^2} \qquad R^w = \frac{\tau^2}{\tau^2 + \sigma_\varepsilon^2}$$

$$\frac{\psi_\mu^2}{\psi_\mu^2 + (\sigma_\omega^2 + \sigma_\varepsilon^2)}$$

Cannot be separated

Discussion

- LMMs and (multilevel) SSMs can be special cases of one another (Piepho & Ogutu, 2007; Tsimikas & Ledolter, 1997).
  - LMM: is easier handling data measured across many participants but with relatively few measurement occasions $N > T$.
  - SSM: is well suited as a tool for representing more complex intraindividual dynamics, particularly when $T > N$.
- Guidelines: if you care about
  - Reliability changes over the time ⇒ LMM
  - Reliability for each person and overall population ⇒ SSM
- Further studies are needed to clarify the properties of these reliability
  - E.g., the role of personal serial fluctuation/within-person variance (Cor vs. VP)

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Q&A

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